

RAG: Chunking Techniques

Introduction

Retrieval-Augmented Generation (RAG) is a framework that enhances large language models (LLMs) by integrating external knowledge retrieval during response generation. Rather than relying solely on the model's internal parameters, RAG retrieves relevant information from a knowledge base (e.g., documents, databases, or web sources) and uses it to inform more accurate, grounded, and up-to-date outputs.

A critical component of the RAG pipeline is **chunking**, the process of dividing large texts or documents into smaller, manageable, and semantically meaningful segments called **chunks**. Effective chunking improves retrieval quality by ensuring that relevant context can be efficiently indexed and surfaced during a query. Poor chunking, on the other hand, can lead to irrelevant or partial context being retrieved, reducing the model's performance.

In this project we explore different techniques, its difference and then we check the performance of each method.

In order to do that, I use a text that is about the women's football in England and how it was banned in 1921. The text was gotten from <https://www.thefa.com/womens-girls-football/heritage/kicking-down-barriers> using Web Scraping.

Before starting

Let's start by reading the file

```
In [24]: lines = []
with open('english_women_football.txt', 'r') as file:
    for line in file:
        if line.strip():
            lines.append(line.strip())

file_lines = "\n".join(lines)
lines
```

Out[24]: ['EARLY YEARS',

'Although the game of football. has been played for centuries - for example, Shakespeare mentions it in two of his plays: The Comedy of Errors (1594) and King Lear (1608) - you might think that women have only been playing over the last generation or so. In fact, their participation dates back much further than you might imagine. It is claimed that Mary Queen of Scots herself owned the oldest football in existence. Sir Philip Sidney's poem A Dialogue Between Two Shepherds, written in 1580, refers to women playing:',

'A tyme there is for all, my mother often sayes,',

'When she, with skirts tuckt very hy, with girles at football playes.'',

'Football remained a popular pastime, but it took another 300 years for the game to be codified into a standard set of rules by the newly formed Football Association in 1863. These rules were partly intended to stop on-pitch violence and make it more acceptable for women to play.',

'The first recorded women's football match was on 7 May 1881, when a supposed 'Scotland' women's team played 'England' at Easter Road, Edinburgh. In actual fact the birthplace and therefore nationalities of the players has been called into question and it's believed at least some of the players were from the theatre community 'acting' out these team roles.',

'The two teams played several matches, but two - in Glasgow and Manchester - were abandoned due to pitch invasions. In Glasgow the players had to escape in a horse-drawn bus! In the first match - which wasn't abandoned - Lily St Clair netted the first goal in front of a crowd of about 2,000 and in doing so became the first recorded female goal scorer in the history of the game. The team representing Scotland were 2-0 winners.',

'THE GAME GROWS',

'During World War One, women increasingly took on traditionally male jobs, particularly and typically working in factories. Sporting activity within these often dark and dirty factories was encouraged to benefit health and wellbeing, as well as to help aid morale and productivity. Football was a natural outlet for this and many factories developed their own teams. In 1917 the munitions factories gave birth to the Munitionettes' Cup, with Blyth Spartans Munitionettes beating Bolckow, Vaughan & Co 5-0 in the final.',

'By far the most famous and one of the most successful of the factory-based teams of the time was Dick, Kerr & Co. From 1917 right through to 1965 they played 800 games - winning 746! They also produced the country's best-ever goal scorer, Lily Parr, who is believed to have scored around 900 goals during her 30-year career. During and after the War, Dick, Kerr Ladies toured the country playing charity games to raise money for injured servicemen. Arguably, they were the first 'professional' women's side and players were paid 10 shillings (50p) to cover their expenses.',

'By 1921 there were about 150 women's football clubs. Matches were popular spectator events and some drew up to 45,000 fans. The future of the women's game looked bright.',

'Despite its growing popularity and the re-emergence of international matches after World War One, controversy was just around the corner. Many factories closed and women went back into domestic life or retrained in professions such as bus conductors and nurses. At the same time, some people questioned whether football was damaging women's health. Dr Mary Scharlieb of Harley Street for example described it as the "most unsuitable game, too much for a women's physical frame".',

'BANNED',

'On 5 December 1921, the FA met at its headquarters in London and announced a ban on the women's game from being played at the professional grounds and pitches of clubs affiliated to the FA, stating "the game of football is quite unsuitable for females and ought not to be encouraged."',

'The ban meant the women's game was side-lined to being played in public parks for

r nearly 50 years.',

'Despite this, women were determined to keep playing football and around 30 teams from across England met in Liverpool on 10 December 1921 to formalise the English Ladies Football Association (ELFA). This aimed "to popularise the game among girls and to assist charity".',

'The following year saw the first and only ELFA Challenge Cup competition. Stoke Ladies lifted the trophy, beating Doncaster and Bentley Ladies 3-1 in the June 1922 final.'

'Though stymied by a lack of decent-sized facilities, women's football carried on throughout the '30s, '40s, '50s and most of the 1960s. Games were played at rugby venues and smaller grounds which couldn't house many fans. Though the women continued to play, they became increasingly overshadowed by the men's game. In 1947, Kent County Football Association suspended a referee because he was also working as manager of the Kent Ladies Football Club. It justified its decision saying that "women's football brings the game into disrepute".',

'But clubs continued to start up, a notable example is Manchester Corinthians Ladies FC founded in 1949. Percy Ashley, a scout for Bolton Wanderers, founded the club so his daughter Doris, a talented footballer, could play in a team. The Corinthians played at Fog Lane Park in Didsbury, a suburb south of the city. Players often had to clean themselves in the local duck pond after matches as there was no running water in the changing rooms. The team were very successful, winning many domestic trophies within their first two years. As there were very few local women's teams, Ashley set up a second team, The Nomads, in 1957 so they could play against the Corinthians in charity matches.'

'The Corinthians went on to win an unofficial European club championship in Germany in 1957. The team was accompanied by Manchester City's German goalkeeper, Bert Trautmann, who acted as their interpreter. More overseas trips followed, including to Portugal/Madeira, the Netherlands, South America and the Caribbean, and Morocco. The team won more than 50 trophies and raised £275,000 for charity.'

'REVIVAL',

'As society began to change during the 1960s, women started to campaign more for their social rights. On 1 November 1969, representatives of 44 clubs attended the first meeting of the Women's Football Association (WFA) in London. Arthur Hobbs, a carpenter and amateur footballer, was the first Honorary Secretary of The WFA and Pat Dunn was its first Chair. In 1967, Hobbs had organised a women's tournament in Deal, Kent, with the support of local miners from Betteshanger Colliery. The Deal Tournament, played on the Colliery's playing fields, showcased and celebrated women's football. It was a pre-cursor to the Women's FA Cup, which would go on to be first played in 1971.'

'In 1972, Hobbs stepped down as Honorary Secretary due to ill health. He was succeeded by Patricia Gregory, who had been instrumental in setting up the WFA. Gregory founded and played for White Ribbon, a team in London who participated in the Deal Tournament. She went on to have a long career in football administration, serving as a member of the UEFA Women's Football Committee from 1980 - 1994.'

'The FA were coming under increasing pressure to lift the ban on women's football. In December 1969, the FA's Executive Committee recommended that what they termed 'ladies' football' should no longer be classified as unaffiliated. In the following month, at a meeting of the FA Council on 19 January 1970, the FA finally voted to rescind the controversial resolution of 1921 that had banned women's football for being "unsuitable".',

"But it took a lot more deliberations, crucially including a meeting of the FA's Rules Revision Committee in August, to prepare the way for an amendment to be made to the relevant FA rule. This couldn't be made until the FA's next Annual General Meeting, which was held at the Great Western Royal Hotel in Paddington on 24 June 1971. Women's football could now be played on FA-affiliated grounds and registered referees could officiate at women's matches.",

'In addition to official league matches, came the launch of a women's club knock-out competition, later to become the Women's FA Cup. In 1971 Southampton won the first final and they would go on to lift the trophy eight times in its first eleven years. In 1971, UEFA recommended that the women's game should be governed by national associations - then numbering 32 - across Europe. The FA made their announcement on recognising The Women's FA (WFA) on 29 February 1972.',

"In November of the same year - 1972 - the first official WFA England team travelled to Scotland for an international match in which England came from behind to win 3-2 with goals from Sylvia Gore, Lynda Hale and Jeannie Allott. That 2-0 defeat between two unofficial sides some 91 years earlier had finally been avenged. The match itself came exactly 100 years after the first recognised men's international, also played between Scotland and England. At this time opportunities for women to represent England in international competitions were still scarce. Between 1972 and 1978, there were only 25 international matches played.",

'Now the push was on to integrate women's football into the fabric of our national sport. That push continued to be led by the WFA. In 1981 they opened their own administrative office although the sport continued to be largely run through the dedication of volunteers. The WFA grew the women's game throughout the 1970s, '80s and early '90s.',

"The FA however was beginning to take a keener interest and in May 1984 the WFA was affiliated with the same status as County FAs. This opened the doors to more central FA support, but much more still needed to be done. That same year, 1984, saw England lose to Sweden - 4-3 on penalties - in the two-legged final of the first UEFA competition for national representative women's teams. This was the predecessor to the Women's EUROs.",

'A DECADE OF FIRSTS',

'The 1990s saw the first strategic plans to develop the women's game in England, together with the first 20 Centres of Excellence for girls and also, in 1998, the first full-time England women's coach, Hope Powell - later to be honoured with both an OBE and CBE.',

'1991 saw another key moment when the WFA launched a 24-club national league which the following year was expanded to three divisions of ten teams (Premier, Division 1 North and Division 1 South). Further fundamental change happened two years later, in 1993, as the WFA voted to pass over its activities to the FA. It no longer had the financial resources to develop the game it had nurtured. The FA established a Women's Football Committee and with it the full-time post of Women's Football Co-ordinator.',

"The 1993-94 season also saw the WFA Cup brought under the control of the FA - 137 teams entered. A year later and the WFA's national league and league cup were also managed by The FA. This saw the birth of the Women's Premier League (FAWPL).",

'FIFA introduced the women's World Cup competition in 1991 and as the women's game started to grow globally, the decade culminated in the finals in the USA in 1999 that featured sold-out stadia and a 90,000 crowd at the final.',

'The game was gaining media attention by the day and more importantly, more participants. By 2002 in England, it had become the top participation team sport for women and girls. This FA target was achieved three years ahead of target. In the same year, Lily Parr of Dick, Kerr Ladies became the first woman inducted into the National Football Museum Hall of Fame. This was in part the game acknowledging its roots, but also building its future foundations.',

'These foundations were given a further boost in 2005 when England hosted the UEFA Women's EURO. Records for crowd attendance and TV audiences were smashed. England beat Finland 2-1 in front of a then European record crowd of 29,092 at the City of Manchester Stadium. Crucially, the talent pool was getting deeper - there were now England teams at various age groups and the senior team - the Lionesses - were beginning to show their talents on the world stage. The 2000s ended with the senior team winning the Cyprus Cup which was their first international trophy and later

in the same year, reaching the final of UEFA Women's EURO 2009, albeit losing out to Germany who were the dominant force at the time. The WU19s however won their UEFA Championship in Belarus.',

"The London 2012 Olympic Games gave women's football in this country a further boost, when the Team GB Women reached the quarter-finals of their competition. Significantly, a 1-0 win over Brazil in Team GB's third group match was played in front of 70,584 fans at Wembley Stadium. This contrasted markedly with the very first England women's international match played at Wembley only 23 years beforehand (England vs Sweden on 23 May 1989) which took place as a 'warm up' before a men's international kick off.",

"As England's international teams were getting ever stronger, club football re-formed itself and 2011 saw the inception of The FA Women's Super League (WSL) which was initially an eight-team summer competition. This replaced the FA Women's Premier League National Division as the highest level of women's football in England and ran on this basis until 2017 when it finally grew into the two-division fully professional game we know today, taking its place alongside the traditional men's professional season, with media interest, spectator levels and sponsorship income having established a solid commercial platform. The top three teams each season qualify for the UEFA Women's Champions League. The WSL has increased the visibility of women's club football across the world, attracting star players from overseas and broadcast partners in Australia, Canada, Dominican Republic, Mexico, Germany, Italy, Scandinavia, New Zealand and the US. Below the two professional divisions, the game's pyramid – the Women's National League – has continued to develop. It now means there are recognised pathways into the women's professional game, with clubs themselves operating Academy structures.",

"In 2024, the Barclays WSL and the second tier Barclays Women's Championship, migrated to become independent of the FA, with the introduction of Women's Professional Leagues Limited with the aim of spearheading a new era in the sport, dedicated to driving growth and maximising potential, placing players and fans at its core.",

'NEW ERA',

'The last 15 years have seen some major developments for both women and girls.',

'In 2013, the FA, Sport England, the Premier League and the Football League Trust launched their first joint national participation programme for girls' football.',

'In 2014, England Women played their first match at the new Wembley Stadium, attracting a then record crowd of 45,619 for their match against Germany. The senior team was by now a serious contender on both European and world stages. They took bronze in the 2015 FIFA Women's World Cup in Canada and reached the semi-finals of UEFA Women's 2017 EURO and got to the same stage again in the FIFA Women's World Cup two years later in France 2019. Their semi-final defeat to the USA attracted a record 11.7m viewers on BBC One.',

'Earlier in the same year they won the SheBelieves Cup for the first time. This success saw ever-increasing crowds at England matches, with 77,786 fans at Wembley to see the senior team face Germany in November 2019. It is little wonder then that Barclays wanted to secure its place as the title partner of The FA WSL in what is believed to be the biggest-ever investment in UK women's sport by a brand.',

'Double participation, double the fanbase and consistent success on the world stage. Those were the three bold goals in The Gameplan for Growth, The FA's first formal strategy for women's and girls' football in England, unveiled in March 2017. All three goals were scored – with more than one million girls (aged 5-15) and 1.9 million women (16+) playing the game in England by the end of the strategy. Numerous FA participation programmes brought girls to football at various age groups and the fan base continued to grow.',

'Following the first three-year strategy, a new one – Inspiring Positive Change – was launched in October 2020. Among its eight goals, one stands out – to give every school-going girl the same access to football as boys, whether at school or in c

lubs. This was followed by Reaching Higher - a new women's and girls' game strategy for 2024-2028.',

'A home win in Wembley for England at the UEFA Women's EURO 2022 in front of a crowd of 87,192 has firmly established the women's game in the national psyche.',

'Media coverage and general interest in the game has never been higher. As well as a huge success on the field, the game-changing tournament created over 400,000 new opportunities for girls and women to engage in grassroots legacy football activities. The win was the glorious icing on the cake of a decade of growth and investment. It was a wonderful serendipity that the win came in the 50th anniversary year of England Women; a moment we all got to celebrate at Wembley on 7 October 2022 when Jill Scott presented the original 1972 team with their long-awaited caps.',

"A year later, the Lionesses made more history in Australia when they reached the FIFA Women's World Cup Final in Sydney, a first-ever appearance at that stage of the competition following an exciting campaign 'down under'. Unfortunately, Sarina Wiegman's team came up against a Spanish team who had swept all before them in the competition and lost by 1-0 in the Final.",

"However, that disappointment was replaced with further celebrations when the Lionesses retained their UEFA Women's EURO crown at the 2025 tournament in Switzerland with a penalty shootout victory over Spain in the final to become the first England team to win a major title on foreign soil."]

Overlapping

Related to chunking I should mention the concept of **overlapping**. Overlapping in chunking means that adjacent chunks share some portion of text (e.g., sentences or tokens). This helps preserve context across chunk boundaries and reduces the chance of cutting off important information.

In overlapping chunking, each new chunk reuses some tokens from the previous chunk. This shared portion (the overlap) ensures smoother context continuity, especially when important information spans across boundaries.

Example (with 50-token chunks and 10-token overlap):

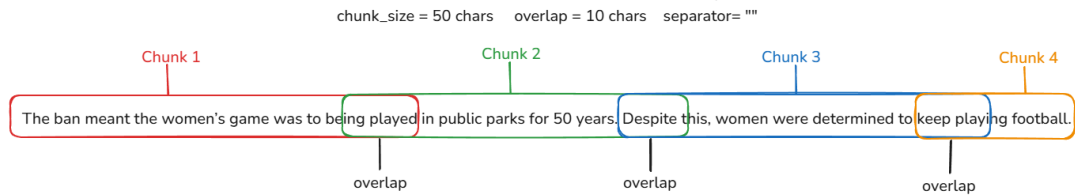
- Chunk 1: Tokens 0–49
- Chunk 2: Tokens 40–89 (overlapping tokens 40–49)
- Chunk 3: Tokens 80–129 (overlapping tokens 80–89)

This way, key information that falls near the edge of a chunk isn't lost between chunks during retrieval.

Chunking Methods

Fixed Size Chunking

Fixed Size Chunking



Fixed-Size Chunking breaks text into chunks of a predefined, fixed size, typically by a number of characters, tokens, or words.

Choose a **chunk size** (e.g. 500 tokens), optionally a fixed **overlap** (e.g. 50 tokens) and a chunk **separator**.

It splits the text into chunks of exactly that size, regardless of sentence, paragraph boundaries or structure of the content.

The separator is used to split the text into pieces first, and then those pieces are grouped into chunks of size $\leq$ chunk_size

Considerations:

- It does not try smaller separators if the chunk is too big.
- It does not preserve semantic or structural boundaries beyond that first split.

How it works:

1. The input text is split using the separator, e.g separator = "\n". Then we have a list of lines.
2. Then it groups these lines into chunks of $\leq$ chunk_size characters, by concatenating multiple lines until the limit is hit.

But if any single line is already longer than chunk_size, It will just include that line as a chunk and it shows a warning like: "Created a chunk of size 518, which is longer than the specified 30".

The chunk size is used to control how many combined short lines go into a chunk. So, **Not to split individual lines that are too long**

Implementation

In order to impment this, I use CharacterTextSplitter from **LangChain**.

```
In [25]: from langchain_text_splitters import CharacterTextSplitter
text_splitter = CharacterTextSplitter()
```

```
chunks = text_splitter.split_text(file_lines)
len(chunks)
```

Out[25]: 1

In [26]: chunks

Out[26]: ['EARLY YEARS\nAlthough the game of football. has been played for centuries – for example, Shakespeare mentions it in two of his plays: The Comedy of Errors (1594) and King Lear (1608) – you might think that women have only been playing over the last generation or so. In fact, their participation dates back much further than you might imagine. It is claimed that Mary Queen of Scots herself owned the oldest football in existence. Sir Philip Sidney’s poem A Dialogue Between Two Shepherds, written in 1580, refers to women playing:\n‘A tyme there is for all, my mother often sayes,\nWhen she, with skirts tuckt very hy, with girles at football playes.’\nFootball remained a popular pastime, but it took another 300 years for the game to be codified into a standard set of rules by the newly formed Football Association in 1863. These rules were partly intended to stop on-pitch violence and make it more acceptable for women to play.\nThe first recorded women’s football match was on 7 May 1881, when a supposed ‘Scotland’ women’s team played ‘England’ at Easter Road, Edinburgh. In actual fact the birthplace and therefore nationalities of the players has been called into question and it’s believed at least some of the players were from the theatre community ‘acting’ out these team roles.\nThe two teams played several matches, but two – in Glasgow and Manchester – were abandoned due to pitch invasions. In Glasgow the players had to escape in a horse-drawn bus! In the first match – which wasn’t abandoned – Lily St Clair netted the first goal in front of a crowd of about 2,000 and in doing so became the first recorded female goal scorer in the history of the game. The team representing Scotland were 2-0 winners.\nTHE GAME GROWS\nDuring World War One, women increasingly took on traditionally male jobs, particularly and typically working in factories. Sporting activity within these often dark and dirty factories was encouraged to benefit health and wellbeing, as well as to help aid morale and productivity. Football was a natural outlet for this and many factories developed their own teams. In 1917 the munitions factories gave birth to the Munitionettes’ Cup, with Blyth Spartans Munitionettes beating Bolckow, Vaughan & Co 5-0 in the final.\nBy far the most famous and one of the most successful of the factory-based teams of the time was Dick, Kerr & Co. From 1917 right through to 1965 they played 800 games – winning 746! They also produced the country’s best-ever goal scorer, Lily Parr, who is believed to have scored around 900 goals during her 30-year career. During and after the War, Dick, Kerr Ladies toured the country playing charity games to raise money for injured servicemen. Arguably, they were the first ‘professional’ women’s side and players were paid 10 shillings (50p) to cover their expenses.\nBy 1921 there were about 150 women’s football clubs. Matches were popular spectator events and some drew up to 45,000 fans. The future of the women’s game looked bright.\nDespite its growing popularity and the re-emergence of international matches after World War One, controversy was just around the corner. Many factories closed and women went back into domestic life or retrained in professions such as bus conductors and nurses. At the same time, some people questioned whether football was damaging women’s health. Dr Mary Sc harlieb of Harley Street for example described it as the “most unsuitable game, too much for a women’s physical frame”. \nBANNED\nOn 5 December 1921, the FA met at its headquarters in London and announced a ban on the women’s game from being played at the professional grounds and pitches of clubs affiliated to the FA, stating “the game of football is quite unsuitable for females and ought not to be encouraged.”\nThe ban meant the women’s game was side-lined to being played in public parks for nearly 50 years.\nDespite this, women were determined to keep playing football and around 30 teams from across England met in Liverpool on 10 December 1921 to formalise the English Ladies Football Association (ELFA). This aimed “to popularise the game among girls and to assist charity”. \nThe following year saw the first and only ELFA Challenge Cup competition. Stoke Ladies lifted the trophy, beating Doncaster and Bentley Ladies 3-1 in the June 1922 final.\nThough stymied by a lack of decent-sized facilities, women’s football carried on throughout the ‘30s, ‘40s, ‘50s and most of the 1960s. Games were played at rugby venues and smaller grounds which couldn’t house many fans. Though the women continued to play, they became in

creasingly overshadowed by the men's game. In 1947, Kent County Football Association suspended a referee because he was also working as manager of the Kent Ladies Football Club. It justified its decision saying that "women's football brings the game into disrepute". But clubs continued to start up, a notable example is Manchester Corinthians Ladies FC founded in 1949. Percy Ashley, a scout for Bolton Wanderers, founded the club so his daughter Doris, a talented footballer, could play in a team. The Corinthians played at Fog Lane Park in Didsbury, a suburb south of the city. Players often had to clean themselves in the local duck pond after matches as there was no running water in the changing rooms. The team were very successful, winning many domestic trophies within their first two years. As there were very few local women's teams, Ashley set up a second team, The Nomads, in 1957 so they could play against the Corinthians in charity matches. The Corinthians went on to win an unofficial European club championship in Germany in 1957. The team was accompanied by Manchester City's German goalkeeper, Bert Trautmann, who acted as their interpreter. More overseas trips followed, including to Portugal/Madeira, the Netherlands, South America and the Caribbean, and Morocco. The team won more than 50 trophies and raised £275,000 for charity.

REVIVAL

As society began to change during the 1960s, women started to campaign more for their social rights. On 1 November 1969, representatives of 44 clubs attended the first meeting of the Women's Football Association (WFA) in London. Arthur Hobbs, a carpenter and amateur footballer, was the first Honorary Secretary of The WFA and Pat Dunn was its first Chair. In 1967, Hobbs had organised a women's tournament in Deal, Kent, with the support of local miners from Betteshanger Colliery. The Deal Tournament, played on the Colliery's playing fields, showcased and celebrated women's football. It was a precursor to the Women's FA Cup, which would go on to be first played in 1971. In 1972, Hobbs stepped down as Honorary Secretary due to ill health. He was succeeded by Patricia Gregory, who had been instrumental in setting up the WFA. Gregory founded and played for White Ribbon, a team in London who participated in the Deal Tournament. She went on to have a long career in football administration, serving as a member of the UEFA Women's Football Committee from 1980 - 1994. The FA were coming under increasing pressure to lift the ban on women's football. In December 1969, the FA's Executive Committee recommended that what they termed 'ladies' football' should no longer be classified as unaffiliated. In the following month, at a meeting of the FA Council on 19 January 1970, the FA finally voted to rescind the controversial resolution of 1921 that had banned women's football for being "unsuitable". But it took a lot more deliberations, crucially including a meeting of the FA's Rules Revision Committee in August, to prepare the way for an amendment to be made to the relevant FA rule. This couldn't be made until the FA's next Annual General Meeting, which was held at the Great Western Royal Hotel in Paddington on 24 June 1971. Women's football could now be played on FA-affiliated grounds and registered referees could officiate at women's matches. In addition to official league matches, came the launch of a women's club knock-out competition, later to become the Women's FA Cup. In 1971 Southampton won the first final and they would go on to lift the trophy eight times in its first eleven years. In 1971, UEFA recommended that the women's game should be governed by national associations - then numbering 32 - across Europe. The FA made their announcement on recognising The Women's FA (WFA) on 29 February 1972. In November of the same year - 1972 - the first official WFA England team travelled to Scotland for an international match in which England came from behind to win 3-2 with goals from Sylvia Gore, Lynda Hale and Jeannie Allott. That 2-0 defeat between two unofficial sides some 91 years earlier had finally been avenged. The match itself came exactly 100 years after the first recognised men's international, also played between Scotland and England. At this time opportunities for women to represent England in international competitions were still scarce. Between 1972 and 1978, there were only 25 international matches played. Now the push was on to integrate women's football into the fabric of our national sport. That push continued to be led by the WFA. In 1981 they opened

their own administrative office although the sport continued to be largely run through the dedication of volunteers. The WFA grew the women's game throughout the 1970s, '80s and early '90s. The FA however was beginning to take a keener interest and in May 1984 the WFA was affiliated with the same status as County FAs. This opened the doors to more central FA support, but much more still needed to be done. That same year, 1984, saw England lose to Sweden - 4-3 on penalties - in the two-legged final of the first UEFA competition for national representative women's teams. This was the predecessor to the Women's EUROs. A DECADE OF FIRSTS The 1990s saw the first strategic plans to develop the women's game in England, together with the first 20 Centres of Excellence for girls and also, in 1998, the first full-time England women's coach, Hope Powell - later to be honoured with both an OBE and CBE. 1991 saw another key moment when the WFA launched a 24-club national league which the following year was expanded to three divisions of ten teams (Premier, Division 1 North and Division 1 South). Further fundamental change happened two years later, in 1993, as the WFA voted to pass over its activities to the FA. It no longer had the financial resources to develop the game it had nurtured. The FA established a Women's Football Committee and with it the full-time post of Women's Football Co-ordinator. The 1993-94 season also saw the WFA Cup brought under the control of the FA - 137 teams entered. A year later and the WFA's national league and league cup were also managed by The FA. This saw the birth of the Women's Premier League (FAWPL). FIFA introduced the women's World Cup competition in 1991 and as the women's game started to grow globally, the decade culminated in the finals in the USA in 1999 that featured sold-out stadia and a 90,000 crowd at the final. The game was gaining media attention by the day and more importantly, more participants. By 2002 in England, it had become the top participation team sport for women and girls. This FA target was achieved three years ahead of target. In the same year, Lily Parr of Dick, Kerr Ladies became the first woman inducted into the National Football Museum Hall of Fame. This was in part the game acknowledging its roots, but also building its future foundations. These foundations were given a further boost in 2005 when England hosted the UEFA Women's EURO. Records for crowd attendance and TV audiences were smashed. England beat Finland 2-1 in front of a then European record crowd of 29,092 at the City of Manchester Stadium. Crucially, the talent pool was getting deeper - there were now England teams at various age groups and the senior team - the Lionesses - were beginning to show their talents on the world stage. The 2000s ended with the senior team winning the Cyprus Cup which was their first international trophy and later in the same year, reaching the final of UEFA Women's EURO 2009, albeit losing out to Germany who were the dominant force at the time. The WU19s however won their UEFA Championship in Belarus. The London 2012 Olympic Games gave women's football in this country a further boost, when the Team GB Women reached the quarter-finals of their competition. Significantly, a 1-0 win over Brazil in Team GB's third group match was played in front of 70,584 fans at Wembley Stadium. This contrasted markedly with the very first England women's international match played at Wembley only 23 years beforehand (England vs Sweden on 23 May 1989) which took place as a 'warm up' before a men's international kick off. As England's international teams were getting ever stronger, club football re-formed itself and 2011 saw the inception of The FA Women's Super League (WSL) which was initially an eight-team summer competition. This replaced the FA Women's Premier League National Division as the highest level of women's football in England and ran on this basis until 2017 when it finally grew into the two-division fully professional game we know today, taking its place alongside the traditional men's professional season, with media interest, spectator levels and sponsorship income having established a solid commercial platform. The top three teams each season qualify for the UEFA Women's Champions League. The WSL has increased the visibility of women's club football across the world, attracting star players from overseas and broadcast partners in Australia, Canada, Dominican Republic, Mexico, Germany, Italy, Scandinavia, New Zealand and the US. Below the two professional divisions

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NEW ERA

The last 15 years have seen some major developments for both women and girls.

In 2013, the FA, Sport England, the Premier League and the Football League Trust launched their first joint national participation programme for girls' football.

In 2014, England Women played their first match at the new Wembley Stadium, attracting a then record crowd of 45,619 for their match against Germany. The senior team was by now a serious contender on both European and world stages. They took bronze in the 2015 FIFA Women's World Cup in Canada and reached the semi-finals of UEFA Women's 2017 EURO and got to the same stage again in the FIFA Women's World Cup two years later in France 2019. Their semi-final defeat to the USA attracted a record 11.7m viewers on BBC One.

Earlier in the same year they won the SheBelieves Cup for the first time. This success saw ever-increasing crowds at England matches, with 77,786 fans at Wembley to see the senior team face Germany in November 2019. It is little wonder then that Barclays wanted to secure its place as the title partner of The FA WSL in what is believed to be the biggest-ever investment in UK women's sport by a brand.

Double participation, double the fanbase and consistent success on the world stage. These were the three bold goals in The Gameplan for Growth, The FA's first formal strategy for women's and girls' football in England, unveiled in March 2017. All three goals were scored – with more than one million girls (aged 5-15) and 1.9 million women (16+) playing the game in England by the end of the strategy. Numerous FA participation programmes brought girls to football at various age groups and the fan base continued to grow.

Following the first three-year strategy, a new one – Inspiring Positive Change – was launched in October 2020. Among its eight goals, one stands out – to give every school-going girl the same access to football as boys, whether at school or in clubs. This was followed by Reaching Higher – a new women's and girls' game strategy for 2024-2028.

A home win in Wembley for England at the UEFA Women's EURO 2022 in front of a crowd of 87,192 has firmly established the women's game in the national psyche.

Media coverage and general interest in the game has never been higher. As well as huge success on the field, the game-changing tournament created over 400,000 new opportunities for girls and women to engage in grassroots legacy football activities. The win was the glorious icing on the cake of a decade of growth and investment. It was a wonderful serendipity that the win came in the 50th anniversary year of England Women; a moment we all got to celebrate at Wembley on 7 October 2022 when Jill Scott presented the original 1972 team with their long-awaited caps.

A year later, the Lionesses made more history in Australia when they reached the FIFA Women's World Cup Final in Sydney, a first-ever appearance at that stage of the competition following an exciting campaign 'down under'. Unfortunately, Sarina Wiegman's team came up against a Spanish team who had swept all before them in the competition and lost by 1-0 in the Final.

However, that disappointment was replaced with further celebrations when the Lionesses retained their UEFA Women's EURO crown at the 2025 tournament in Switzerland with a penalty shootout victory over Spain in the final to become the first England team to win a major title on foreign soil.']

By default the separator is "\n\n", so we have just one chunk with all the lines. As it was explained, the separator is used to split the text into pieces first, and as it is higher than the chunk size it keeps as it.

Let's set a new separator without overlapping.

```
In [27]: text_splitter = CharacterTextSplitter(chunk_size=50, chunk_overlap=0, separator=" "
chunks = text_splitter.split_text(file_lines)
len(chunks)
```

Out[27]: 369

```
In [28]: chunks[0:5]
```

```
Out[28]: ['EARLY YEARS\nAlthough the game of football. has',
'been played for centuries - for example,',
'Shakespeare mentions it in two of his plays: The',
'Comedy of Errors (1594) and King Lear (1608) - you',
'might think that women have only been playing over']
```

With this configuration we have 369 chunks of 50 characters. As we can see, this technique splits in the middle of sentences or words, and doesn't keep the semantic context.

In the first chunk we can see the title and the first line together.

Now let's set a overlapping

```
In [29]: text_splitter = CharacterTextSplitter(chunk_size=50, chunk_overlap=5, separator=" "
chunks = text_splitter.split_text(file_lines)
len(chunks)
```

Out[29]: 394

```
In [32]: chunks[0:10]
```

```
Out[32]: ['EARLY YEARS\nAlthough the game of football. has',
'has been played for centuries - for example,',
'Shakespeare mentions it in two of his plays: The',
'The Comedy of Errors (1594) and King Lear (1608) -',
'- you might think that women have only been',
'been playing over the last generation or so. In',
'In fact, their participation dates back much',
'much further than you might imagine. It is claimed',
'that Mary Queen of Scots herself owned the oldest',
'football in existence. Sir Philip Sidney's poem A']
```

From above we can see that we have more chunks than before, and also how each chunk has words from the previous one.

To summarize this method:

Pros:

- Simple and fast to implement.
- Works well when structure doesn't matter.
- Easy to control memory usage and token limits.

Cons:

- Can split in the middle of sentences or words.
- May lose semantic context if a thought spans across chunks. This doesn't guarantee sentence or paragraph boundaries

As solution of these cons, we have the **Recursive Chunking** method.

Recursive Chunking



Recursive Chunking breaks text into chunks by recursively trying to preserve natural structure (e.g. paragraphs, sentences, words) while staying under a token limit.

How it works:

1. Define a max token size (e.g. 500 tokens).
2. Try splitting by paragraphs first. If a paragraph is too large, then it splits into sentences.
3. If a sentence is still too large, split it into phrases or words.
4. Optionally use overlap between chunks for context retention.

Pros:

- Maintains semantic coherence.
- Avoids splitting mid-sentence or mid-thought (as the previous technique)
- Better for LLMs and semantic search tasks.

Cons:

- More complex to implement.
- Slightly slower due to the recursive process.

Implementation

```
In [34]: from langchain_text_splitters import RecursiveCharacterTextSplitter
text_splitter = RecursiveCharacterTextSplitter(chunk_size=50, chunk_overlap=15)
chunks = text_splitter.split_text(file_lines)
```

```
In [36]: chunks[0:5]
```

```
Out[36]: ['EARLY YEARS',  
          'Although the game of football. has been played',  
          'been played for centuries - for example,',  
          '- for example, Shakespeare mentions it in two of',  
          'it in two of his plays: The Comedy of Errors']
```

We can see how the chunks maintain the semantic of the text, the title is in a new line, instead of all in the same line like in the Fixed Size where we had "EARLY YEARS\nAlthough the game of football. has been"

Here the separator is a list, where the default value is ["\n\n", "\n", " ", ""].

It tries to split first by paragraph, then by new line, then by space and finally by character, until the chunks are small enough.

Semantic Splitting



This means dividing text into chunks based on its meaning, not just character counts, tokens, or punctuation. The idea is to keep conceptually complete and coherent pieces of text together in each chunk. Things like full sentences, paragraphs, or topics — so that LLMs can better understand and work with them

How it Works:

1. Split into candidate units: Usually sentences or paragraphs.
2. Embed each unit: Convert text into vectors using an embedding model (like OpenAI, HuggingFace).
3. Group semantically similar units:
4. Use cosine similarity between adjacent vectors.
5. Merge sentences or paragraphs until the similarity drops or a token limit is hit.
6. Create semantically meaningful chunks.

Pros:

- Preserves meaning: Keeps related sentences and ideas together, reducing loss of context.
- Avoids arbitrary cuts: Doesn't split mid-sentence or mid-thought, unlike fixed-size splitting.
- More human-like: Mimics how humans chunk text — by concepts and ideas.

- Reduces hallucinations: When input is semantically coherent, LLMs are less likely to invent false connections.
- Improves retrieval precision: Better matches for semantic search and vector similarity.

Cons:

- More complex to implement: Requires embeddings, similarity calculations, sometimes clustering algorithms.
- Slower processing: You need to compute sentence embeddings (often in batch), which takes time.
- Chunk sizes can be uneven: No strict control over token/character counts — might produce very large or very small chunks.
- Needs a semantic model: Requires an embedding model (like sentence-transformers or OpenAI's embedding API), which might not be free.

Implementation

```
In [61]: from langchain_experimental.text_splitter import SemanticChunker
from langchain_openai.embeddings import OpenAIEmbeddings
text_splitter = SemanticChunker(OpenAIEmbeddings())
docs = text_splitter.split_text(file_lines)
```

```
In [62]: docs[0:5]
```


Out[62]: ['EARLY YEARS\nAlthough the game of football. has been played for centuries - for example, Shakespeare mentions it in two of his plays: The Comedy of Errors (1594) and King Lear (1608) - you might think that women have only been playing over the last generation or so. In fact, their participation dates back much further than you might imagine. It is claimed that Mary Queen of Scots herself owned the oldest football in existence. Sir Philip Sidney's poem A Dialogue Between Two Shepherds, written in 1580, refers to women playing:\n'A tyme there is for all, my mother often sayes,\nWhen she, with skirts tuckt very hy, with girles at football playes. '\nFootball remained a popular pastime, but it took another 300 years for the game to be codified into a standard set of rules by the newly formed Football Association in 1863. These rules were partly intended to stop on-pitch violence and make it more acceptable for women to play.',

'The first recorded women's football match was on 7 May 1881, when a supposed 'Scotland' women's team played 'England' at Easter Road, Edinburgh. In actual fact the birthplace and therefore nationalities of the players has been called into question and it's believed at least some of the players were from the theatre community 'acting' out these team roles.',

'The two teams played several matches, but two - in Glasgow and Manchester - were abandoned due to pitch invasions.',

'In Glasgow the players had to escape in a horse-drawn bus! In the first match - which wasn't abandoned - Lily St Clair netted the first goal in front of a crowd of about 2,000 and in doing so became the first recorded female goal scorer in the history of the game. The team representing Scotland were 2-0 winners. THE GAME GROWS\nDuring World War One, women increasingly took on traditionally male jobs, particularly and typically working in factories. Sporting activity within these often dark and dirty factories was encouraged to benefit health and wellbeing, as well as to help aid morale and productivity. Football was a natural outlet for this and many factories developed their own teams. In 1917 the munitions factories gave birth to the Munitionettes' Cup, with Blyth Spartans Munitionettes beating Bolckow, Vaughan & Co 5-0 in the final. By far the most famous and one of the most successful of the factory-based teams of the time was Dick, Kerr & Co. From 1917 right through to 1965 they played 800 games - winning 746! They also produced the country's best-ever goal scorer, Lily Parr, who is believed to have scored around 900 goals during her 30-year career. During and after the War, Dick, Kerr Ladies toured the country playing charity games to raise money for injured servicemen. Arguably, they were the first 'professional' women's side and players were paid 10 shillings (50p) to cover their expenses. By 1921 there were about 150 women's football clubs. Matches were popular spectator events and some drew up to 45,000 fans. The future of the women's game looked bright. Despite its growing popularity and the re-emergence of international matches after World War One, controversy was just around the corner. Many factories closed and women went back into domestic life or retrained in professions such as bus conductors and nurses. At the same time, some people questioned whether football was damaging women's health. Dr Mary Scharlieb of Harley Street for example described it as the "most unsuitable game, too much for a woman's physical frame". BANNED\nOn 5 December 1921, the FA met at its headquarters in London and announced a ban on the women's game from being played at the professional grounds and pitches of clubs affiliated to the FA, stating "the game of football is quite unsuitable for females and ought not to be encouraged." \nThe ban meant the women's game was side-lined to being played in public parks for nearly 50 years. Despite this, women were determined to keep playing football and around 30 teams from across England met in Liverpool on 10 December 1921 to formalise the English Ladies Football Association (ELFA). This aimed "to popularise the game among girls and to assist charity". The following year saw the first and only ELFA Challenge Cup competition. Stoke Ladies lifted the trophy, beating Doncaster and Bentley Ladies 3-1 in the June 1922 final. Though stymied by a lack of decent-sized facilities, women's football carried on throughout the '30s, '40s, '50s and most of the 1

960s. Games were played at rugby venues and smaller grounds which couldn't house many fans. Though the women continued to play, they became increasingly overshadowed by the men's game. In 1947, Kent County Football Association suspended a referee because he was also working as manager of the Kent Ladies Football Club. It justified its decision saying that "women's football brings the game into disrepute". But clubs continued to start up, a notable example is Manchester Corinthians Ladies FC founded in 1949. Percy Ashley, a scout for Bolton Wanderers, founded the club so his daughter Doris, a talented footballer, could play in a team. The Corinthians played at Fog Lane Park in Didsbury, a suburb south of the city. Players often had to clean themselves in the local duck pond after matches as there was no running water in the changing rooms.',

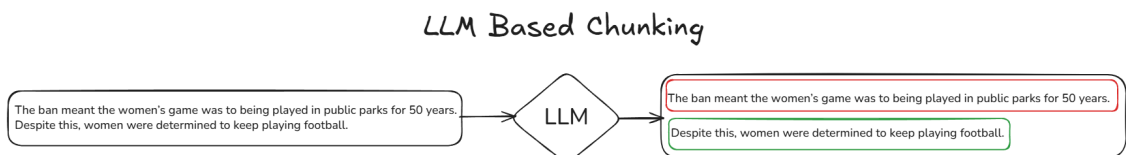
'The team were very successful, winning many domestic trophies within their first two years. As there were very few local women's teams, Ashley set up a second team, The Nomads, in 1957 so they could play against the Corinthians in charity matches. The Corinthians went on to win an unofficial European club championship in Germany in 1957. The team was accompanied by Manchester City's German goalkeeper, Bert Trautmann, who acted as their interpreter. More overseas trips followed, including to Portugal/Madeira, the Netherlands, South America and the Caribbean, and Morocco.'

```
In [63]: sizes = [len(doc) for doc in docs]
        sizes
```

```
Out[63]: [929, 357, 114, 3881, 576, 1150, 4468, 6214]
```

This has generated 8 chunks, all with different sizes.

LLM Based Chunking



This method uses a LLM to generate chunks. The LLM processes the text and generates semantically isolated sentences or propositions that can stand alone.

This is different from:

- Fixed-size splitting (blindly slicing by tokens)
- Recursive splitting (splitting by separators)
- Embedding-based splitting (using vector similarity)

Here, the LLM uses its own understanding to decide how to split.

Pros:

- Extremely flexible (you can tell the LLM exactly how to split, example to take care of the title, paragraph..)
- Very semantically aware

- No need to hand-craft separator rules

Cons:

- Slower (LLM inference per document)
- Costlier (uses tokens to process the entire text, and could lead to context overflow)
- Needs good prompt engineering

What is the difference between this and Semantic Chunking?

- This uses a LLM to segment the text. On the other hand the Semantic uses Embeddings.
- The LLM reads the full text and intelligently splits it into coherent, meaningful chunks. On the other hand Semantic uses embedding similarity to decide where to split or merge based on semantic shifts.
- LLM is slower and more expensive, Semantic is faster and cheaper (uses embedding model only)

LLM is like asking a smart editor to break your document into logical sections and Semantic is like a machine looking for topic changes by comparing paragraph meanings

Implementation

```
In [ ]: from langchain_openai import ChatOpenAI
        from langchain_core.prompts import ChatPromptTemplate
        import json

        prompt = ChatPromptTemplate.from_template("""
        You are an expert document processor.
        Split the following text into coherent chunks of **no more than {chunk_size} characters.
        Try to preserve sentence boundaries, but keep the chunk under {chunk_size} characters.
        Each chunk should be semantically meaningful and should not split sentences mid-way.

        Return the result as a JSON array of strings, like this:
        [
            "Chunk 1 text here...",
            "Chunk 2 text here...",
            ...
        ]
        Only return the JSON. Do not include any other text.

        Text:
        {text}
        """)

        llm = ChatOpenAI(model="gpt-3.5-turbo", temperature=0)
        chain = prompt | llm
        response = chain.invoke({
            "chunk_size": 50,
            "text": file_lines
        })
```

```
chunks = json.loads(response.content)
```

```
In [ ]: chunks[0:5]
```

```
Out[ ]: ['EARLY YEARS Although the game of football has been played for centuries – for example, Shakespeare mentions it in two of his plays: The Comedy of Errors (1594) and King Lear (1608) – you might think that women have only been playing over the last generation or so.',  
        'In fact, their participation dates back much further than you might imagine. It is claimed that Mary Queen of Scots herself owned the oldest football in existence. Sir Philip Sidney’s poem A Dialogue Between Two Shepherds, written in 1580, refers to women playing: ‘A tyme there is for all, my mother often sayes, When she, with skirts tuckt very hy, with girles at football playes.’',  
        'Football remained a popular pastime, but it took another 300 years for the game to be codified into a standard set of rules by the newly formed Football Association in 1863. These rules were partly intended to stop on-pitch violence and make it more acceptable for women to play.',  
        'The first recorded women’s football match was on 7 May 1881, when a supposed ‘Scotland’ women’s team played ‘England’ at Easter Road, Edinburgh. In actual fact the birthplace and therefore nationalities of the players has been called into question and it’s believed at least some of the players were from the theatre community ‘acting’ out these team roles.',  
        'The two teams played several matches, but two – in Glasgow and Manchester – were abandoned due to pitch invasions. In Glasgow the players had to escape in a horse-drawn bus! In the first match – which wasn’t abandoned – Lily St Clair netted the first goal in front of a crowd of about 2,000 and in doing so became the first recorded female goal scorer in the history of the game. The team representing Scotland were 2-0 winners.']
```

```
In [95]: class LLMChunker:  
        def __init__(self, model="gpt-3.5-turbo", chunk_size=50):  
            """  
            Initializes the LLMChunker with the specified model and chunk size.  
            """  
            self.llm = ChatOpenAI(model=model, temperature=0)  
            self.chunk_size = chunk_size  
            self.prompt = ChatPromptTemplate.from_template("""  
            You are an expert document processor.  
            Split the following text into coherent chunks of **no more than {chunk_size}  
            Try to preserve sentence boundaries, but keep the chunk under {chunk_size}  
            Each chunk should be semantically meaningful and should not split sentences  
  
            Return the result as a JSON array of strings, like this:  
            [  
            "Chunk 1 text here...",  
            "Chunk 2 text here...",  
            ...  
            ]  
            Only return the JSON. Do not include any other text.  
  
            Text:  
            {text}  
            """)  
  
        def split_text(self, text):
```

```

"""
Splits the given text into chunks using the LLM.

Args:
    text (str): The text to split.

Returns:
    list: A list of chunks.
"""
chain = self.prompt | self.llm
response = chain.invoke({
    "chunk_size": self.chunk_size,
    "text": text
})
return json.loads(response.content)

```

Chunking Evaluation

In this section, we evaluate the quality of the chunks generated by different chunking methods using predefined metrics. These metrics assess the factual accuracy, relevance to the query, and the contextual usefulness of the retrieved chunks. The evaluation is conducted by generating questions and comparing the performance of each method.

After getting the chunks, how do we know if they are good enough or useful? There are some metrics to validate that.

Some metrics that we can validate are:

- Faithfulness (not hallucinated): Ensures that the retrieved chunks provide accurate information to the model, reducing hallucinations.
- Relevancy (Relevant to the query): Measures how well the retrieved chunks align with the user's query.
- Context Relevancy (Contextually useful to the LLM): Evaluates whether the retrieved chunks provide useful context for generating high-quality answers.

These are metrics/tools used to evaluate the quality of answers in RAG pipelines, especially how well the chunks retrieved contribute to:

Evaluator	Purpose
FaithfulnessEvaluator	Measures truthfulness of the answer vs context
RelevancyEvaluator	Measures how relevant the answer is to the query
ContextRelevancyEvaluator	Measures how relevant the context (chunks) is to the query

Where:

- Context, refers to the chunks retrieved from documents to answer that query
- Query, is the user's question.

- Answer, is the response generated by the LLM using the context and the query

Effective chunking is essential for ensuring that the RAG pipeline retrieves meaningful and accurate information. Poor chunking can result in irrelevant or incomplete context, reducing the quality of the generated responses

Faithfulness Evaluator

What this checks:

- **Is the answer factually supported by the retrieved context?**

Tells you if your chunks provided enough or accurate information to answer the query.

This helps **detect hallucinations** (situations where the model generates answers not grounded in the source chunks)

Useful when comparing different chunking strategies to see which reduces hallucinations.

Relevancy Evaluator

What this checks:

- **Is the answer actually relevant to the query?**

Helps you see if the answer reflects a meaningful match between query intent and retrieved text.

High relevancy = chunks aligned well with the query.

Context Relevancy Evaluator

What this checks:

- **Are the retrieved chunks (context) actually relevant to the query?**

This doesn't look at the final answer, it evaluates just the retrieved context.

Helps evaluate your retrieval and chunking strategy directly.

Tells you: "Did I retrieve good chunks?"

If low, your chunking may be too coarse, fine, or not semantically aligned..

Summary table

Evaluator	Evaluates...	Helps improve...	Common Failure Reason
FaithfulnessEvaluator	Answer vs. context (truth)	Chunk coverage + accuracy	Chunk too small or off-topic
RelevancyEvaluator	Answer vs. query (focus)	Model's answer quality	Irrelevant response
ContextRelevancyEvaluator	Context vs. query (retrieval)	Chunking + retrieval strategy	Wrong or noisy chunks

Chunks Evaluation

1. Chunks Preparations

Before evaluating the performance of different chunking methods, it is essential to prepare the chunks in a structured and consistent manner. This involves splitting the text into chunks using various chunking techniques and configurations, saving these chunks for later use, and ensuring they are ready for evaluation.

In this section, we will:

1. Define the chunking methods and their parameters (e.g., chunk size, overlap).
2. Generate chunks for each method and configuration.
3. Save the prepared chunks to files for efficient retrieval during the evaluation process.

This preparation step ensures that all chunking methods are applied uniformly, enabling a fair comparison of their effectiveness in the subsequent evaluation phase.

```
In [99]: import pickle

def save_chunks(text, chunk_sizes, chunk_path, splitter_class, **splitter_kwargs):
    """
    Splits text into chunks using the provided splitter class and saves them to files.

    Args:
        text (string): The text to split.
        chunk_sizes (list): A list of chunk sizes to use for splitting (ignored for
        chunk_path (str): Path where the files will be saved.
        splitter_class (class): The text splitter class to use (e.g., CharacterTextSplitter).
        **splitter_kwargs: Additional keyword arguments for the splitter class.
    """
    for size in chunk_sizes:
        # Initialize the splitter
        if splitter_class.__name__ == "CharacterTextSplitter":
            splitter = splitter_class(
                chunk_size=size,
                chunk_overlap=size / 10, # Example: 10% overlap
                **splitter_kwargs
            )
```

```

elif splitter_class.__name__ == "SemanticChunker":
    splitter = splitter_class(OpenAIEmbeddings())
elif splitter_class.__name__ == "LLMChunker":
    splitter = splitter_class(chunk_size=50)
elif splitter_class.__name__ == "RecursiveCharacterTextSplitter":
    splitter = splitter_class(
        chunk_size=size,
        chunk_overlap=size / 10,
        **splitter_kwargs
    )

chunks = splitter.split_text(text)
file_name = f"{splitter_class.__name__}_chunks_{size}.pkl"
file_path = chunk_path + file_name

# Save the chunks to a file
with open(file_path, "wb") as file:
    pickle.dump(chunks, file)

print(f"Chunks of size {size} saved to {file_path}")

```

```

In [ ]: chunk_sizes = [50, 100, 150]
        chunk_path = "./chunks/"
        save_chunks(file_lines, chunk_sizes, chunk_path, CharacterTextSplitter, separator='

```

Chunks of size 50 saved to ./chunks/CharacterTextSplitter_chunksize_50.pkl
 Chunks of size 100 saved to ./chunks/CharacterTextSplitter_chunksize_100.pkl
 Chunks of size 150 saved to ./chunks/CharacterTextSplitter_chunksize_150.pkl

```

In [ ]: chunk_sizes = [50, 100, 150]
        chunk_path = "./chunks/"
        save_chunks(file_lines, chunk_sizes, chunk_path, RecursiveCharacterTextSplitter)

```

Chunks of size 50 saved to ./chunks/RecursiveCharacterTextSplitter_chunksize_50.pkl
 Chunks of size 100 saved to ./chunks/RecursiveCharacterTextSplitter_chunksize_100.pkl
 1
 Chunks of size 150 saved to ./chunks/RecursiveCharacterTextSplitter_chunksize_150.pkl
 1

```

In [ ]: chunk_sizes = [1]
        chunk_path = "./chunks/"
        save_chunks(file_lines, chunk_sizes, chunk_path, SemanticChunker)

```

Chunks of size 1 saved to ./chunks/SemanticChunker_chunksize_1.pkl

```

In [100... chunk_sizes = [50]
            chunk_path = "./chunks/"
            save_chunks(file_lines.replace("\n", " "), chunk_sizes, chunk_path, LLMChunker)

```

Chunks of size 50 saved to ./chunks/LLMChunker_chunksize_50.pkl

Let's see now some samples of the chunks per method

```

In [238... import pickle
            import pandas as pd

            methods = [

```



```

    "CharacterTextSplitter_chunksize_50.pkl",
    "RecursiveCharacterTextSplitter_chunksize_100.pkl",
    "SemanticChunker_chunksize_1.pkl",
    "LLMChunker_chunksize_50.pkl"
]
chunk_path = "./chunks/"
sample_chunks = {}

for method in methods:
    with open(chunk_path + method, "rb") as f:
        chunks = pickle.load(f)
        sample_chunks[method.replace(".pkl", "")] = chunks[:3]

df_samples = pd.DataFrame(sample_chunks)
df_samples.index = [f"Chunk {i+1}" for i in range(3)]
df_samples

```

Out[238]...

	CharacterTextSplitter_chunksize_50	RecursiveCharacterTextSplitter_chunksize_100	Ser
Chunk 1	EARLY YEARS\nAlthough the game of football. ha...		EARLY YEARS
Chunk 2	s been played for centuries – for example, Sha...	Although the game of football. has been played...	
Chunk 3	akespeare mentions it in two of his plays: The...	it in two of his plays: The Comedy of Errors (...)	I

Now that we have the methods, I wil ask to a model to create a set of questions from the original text, that will be used to evaluated what is the best split method.

```

In [48]: from langchain_openai import ChatOpenAI
from langchain.prompts import PromptTemplate
translation_llm = ChatOpenAI(model="gpt-3.5-turbo", temperature=0)
translation_prompt = PromptTemplate.from_template(
    """Based on the bellow context, Given the context information and not p
    generate only 40 questions.
    Generate the questions in English.
    Context: {context}"""
)
translated_question = translation_llm.invoke(translation_prompt.format(context="\n"

```

Generated questions:

```

In [51]: translated_question.content.split("\n")

```

```
Out[51]: ["1. When was the first recorded women's football match?",
'2. Who owned the oldest football in existence according to claims?',
'3. What was the name of the poem written by Sir Philip Sidney in 1580 that refer
s to women playing football?',
'4. How many goals did Lily St Clair score in the first match representing Scotla
nd?',
"5. What was the name of the team that won the Munitionettes' Cup in 1917?",
'6. How many games did Dick, Kerr & Co win out of the 800 they played?',
'7. Who is believed to have scored around 900 goals during her 30-year career?',
"8. How many women's football clubs were there in 1921?",
'9. Who described football as "the most unsuitable game, too much for a women\'s
physical frame"?',
"10. When was the ban on women's football lifted by the FA?",
'11. When was the English Ladies Football Association (ELFA) formalized?',
'12. Who won the first ELFA Challenge Cup competition in June 1922?',
"13. When did the FA finally vote to rescind the ban on women's football?",
"14. Who was the first Honorary Secretary of the Women's Football Association (WF
A)?",
"15. When was the first Women's FA Cup played?",
"16. Who won the first Women's FA Cup final in 1971?",
"17. When did the FA recommend that the women's game should be governed by nation
al associations across Europe?",
'18. When did the first official WFA England team play an international match in
Scotland?',
'19. Who scored the winning goal for Team GB Women against Brazil at Wembley Stad
ium in 2012?',
"20. When was the FA Women's Super League (WSL) launched?",
"21. Who was the first full-time England women's coach in 1998?",
'22. When did the WFA vote to pass over its activities to the FA?',
"23. Who won the first UEFA competition for national representative women's teams
in 1984?",
"24. When did the FA Women's Super League grow into a two-division fully professi
onal game?",
'25. Who was the first woman inducted into the National Football Museum Hall of F
ame in 2002?',
'26. When did England Women play their first match at the new Wembley Stadium?',
'27. Who won the SheBelieves Cup for the first time in 2019?',
"28. When was the first joint national participation programme for girls' footbal
l launched?",
"29. Who was the first Chair of the Women's Football Association (WFA)?",
"30. When did the FA launch their first formal strategy for women's and girls' fo
otball in England?",
'31. Who presented the original 1972 team with their long-awaited caps in 2022?',
"32. When did the Lionesses reach the FIFA Women's World Cup Final for the first
time?",
"33. Who did the Lionesses lose to in the FIFA Women's World Cup Final in Sydne
y?",
"34. When did the Lionesses retain their UEFA Women's EURO crown?",
"35. Who did the Lionesses defeat in the UEFA Women's EURO final to retain their
title?",
"36. When was the UEFA Women's EURO 2022 held?",
"37. How many new opportunities for girls and women were created by the UEFA Wome
n's EURO 2022?",
"38. When was the Women's Professional Leagues Limited introduced?",
'39. Who was the first England team to win a major title on foreign soil?']
```

"40. When was the Reaching Higher strategy launched for the women's and girls' game?"]

```
In [ ]: import pandas as pd
# Create a DataFrame of the questions in order to save them to a CSV file
df = pd.DataFrame(translated_question.content.split("\n"), columns=['Questions'])
df.to_csv('questions.csv', index=False)
```

Now that we have the 40 questions, I evaluate the splits methods.

```
In [ ]: from llama_index.core.evaluation import (
    FaithfulnessEvaluator,
    RelevancyEvaluator,
    ContextRelevancyEvaluator
)
from llama_index.core import (
    VectorStoreIndex,
    Document,
)
from llama_index.llms.openai import OpenAI
from llama_index.core import Settings
from llama_index.embeddings.openai import OpenAIEmbedding
import time

model = OpenAI(model="gpt-3.5-turbo", temperature=0)
faithfulness_evaluator = FaithfulnessEvaluator(llm=model)
relevancy_evaluator = RelevancyEvaluator(llm=model)
context_relevancy_evaluator = ContextRelevancyEvaluator(llm=model)
```

```
In [ ]: def evaluate_response_time_and_accuracy(eval_documents, evaluation_questions):
    total_response_time = 0
    total_faithfulness = 0
    total_relevancy = 0
    total_contextrelevancy = 0.0

    Settings.llm = OpenAI(model="gpt-3.5-turbo")
    Settings.embed_model = OpenAIEmbedding()
    # create vector index
    vector_index = VectorStoreIndex.from_documents(
        eval_documents, embed_model=Settings.embed_model
    )
    num_questions = len(evaluation_questions)
    df = []

    for question in evaluation_questions:
        start_time = time.time()
        response_vector = vector_index.as_query_engine(llm=Settings.llm).query(question)
        elapsed_time = time.time() - start_time

        faithfulness_result = faithfulness_evaluator.evaluate_response(
            response=response_vector
        ).passing
        relevancy_result = relevancy_evaluator.evaluate_response(
            query=question, response=response_vector
        ).passing
```

```

context_relevancy_result = context_relevancy_evaluator.evaluate_response(qu

total_response_time += elapsed_time
total_faithfulness += faithfulness_result
total_relevancy += relevancy_result
total_context_relevancy += context_relevancy_result.score

data = {'question': question,
        'response': response_vector.response,
        'context': [{'text': c.text, 'metadata': c.metadata} for c in respo
        'faithfulness': faithfulness_result,
        'relevancy': relevancy_result,
        'context_relevancy_score': context_relevancy_result.score,
        'context_relevancy_feedback': context_relevancy_result.feedback,
        'response_time': elapsed_time}

df.append(data.copy())

average_response_time = total_response_time / num_questions
average_faithfulness = total_faithfulness / num_questions
average_relevancy = total_relevancy / num_questions
average_context_relevancy = total_context_relevancy / num_questions
return average_response_time, average_faithfulness, average_relevancy, average_

```

```

In [ ]: import os
import pandas as pd
import pickle

df = pd.read_csv('questions.csv')
evaluation_questions = df['Questions'].tolist()
results = {}

directory = './chunks'
for filename in os.listdir(directory):
    file_path = os.path.join(directory, filename)
    print(f"Processing file: {file_path}")
    with open(file_path, "rb") as file:
        chunk_docs = pickle.load(file)
    # Documents to llama-index docs
    llama_documents = [Document(text=doc) for doc in chunk_docs]
    avg_response_time, avg_faithfulness, avg_relevancy, average_context_relevancy, d
    results[filename] = { "Average response time": avg_response_time, "Average fait
    file_path = os.path.join('chunks_evaluations', filename.replace('.pkl', '_evalu
    with open(file_path, 'w', encoding='utf-8') as file:
        json.dump(df, file, ensure_ascii=False, indent=4)

file_path = os.path.join('chunks_metrics.json')
with open(file_path, 'w') as file:
    json.dump(results, file)

```

```

Processing file: ./chunks\CharacterTextSplitter_chunksize_100.pkl
Processing file: ./chunks\CharacterTextSplitter_chunksize_150.pkl
Processing file: ./chunks\CharacterTextSplitter_chunksize_50.pkl
Processing file: ./chunks\RecursiveCharacterTextSplitter_chunksize_100.pkl
Processing file: ./chunks\RecursiveCharacterTextSplitter_chunksize_150.pkl
Processing file: ./chunks\RecursiveCharacterTextSplitter_chunksize_50.pkl
Processing file: ./chunks\SemanticChunker_chunksize_1.pkl

```

```

In [ ]: with open('./chunks_metrics.json', 'r') as file:
        metrics = json.load(file)
        df = pd.DataFrame(metrics).T

```

Here we can see the metrics for each method:

```
In [16]: df
```

```
Out[16]:
```

	Average response time	Average faithfulness	Average relevancy	Average context relevancy
CharacterTextSplitter_chunksize_100.pkl	1.373114	0.625	0.450	0.675000
CharacterTextSplitter_chunksize_150.pkl	1.373890	0.625	0.450	0.678125
CharacterTextSplitter_chunksize_50.pkl	1.682769	0.575	0.325	0.590625
RecursiveCharacterTextSplitter_chunksize_100.pkl	1.262391	0.675	0.400	0.718750
RecursiveCharacterTextSplitter_chunksize_150.pkl	1.038087	0.700	0.375	0.678125
RecursiveCharacterTextSplitter_chunksize_50.pkl	0.993778	0.625	0.325	0.584375
SemanticChunker_chunksize_1.pkl	1.164329	0.750	0.400	0.712500

But in order to them better, let's move the data to tha plot, to take better insights.

```

In [183... import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

metrics_df = pd.DataFrame(metrics).T.reset_index().rename(columns={"index": "Chunking Method"})
metrics_df['Chunking Method'] = metrics_df['Chunking Method'].str.replace('.pkl', '')
metrics_to_plot = ['Average response time', 'Average faithfulness', 'Average relevancy']

melted_df = metrics_df.melt(id_vars=["Chunking Method"],
                           value_vars=metrics_to_plot,
                           var_name="Metric",
                           value_name="Score")

# Normalize the scores for opacity adjustment
def normalize_scores(row):
    if row['Metric'] == 'Average response time':
        # Invert normalization for response time (Lower is better)
        return 1 - ((row['Score'] - melted_df[melted_df['Metric'] == 'Average response time']['Score'].min()) /
                    (melted_df[melted_df['Metric'] == 'Average response time']['Score'].max() - melted_df[melted_df['Metric'] == 'Average response time']['Score'].min()))
    else:
        return (row['Score'] - melted_df[melted_df['Metric'] == row['Metric']]['Score'].min()) /
               (melted_df[melted_df['Metric'] == row['Metric']]['Score'].max() - melted_df[melted_df['Metric'] == row['Metric']]['Score'].min())

```

```

else:
    # Normal normalization for other metrics
    return (row['Score'] - melted_df[melted_df['Metric'] == row['Metric']]['Score'].min() -
            (melted_df[melted_df['Metric'] == row['Metric']]['Score'].max() -
             melted_df[melted_df['Metric'] == row['Metric']]['Score'].min()))

melted_df['Normalized Score'] = melted_df.apply(normalize_scores, axis=1)
melted_df['Adjusted Opacity'] = melted_df['Normalized Score'] * 0.9 + 0.1

# Define a unique color for each metric
metric_colors = {
    'Average response time': '#FF5733', # Red
    'Average faithfulness': '#33FF57', # Green
    'Average relevancy': '#3357FF', # Blue
    'Average context relevancy': '#FF33A1' # Pink
}

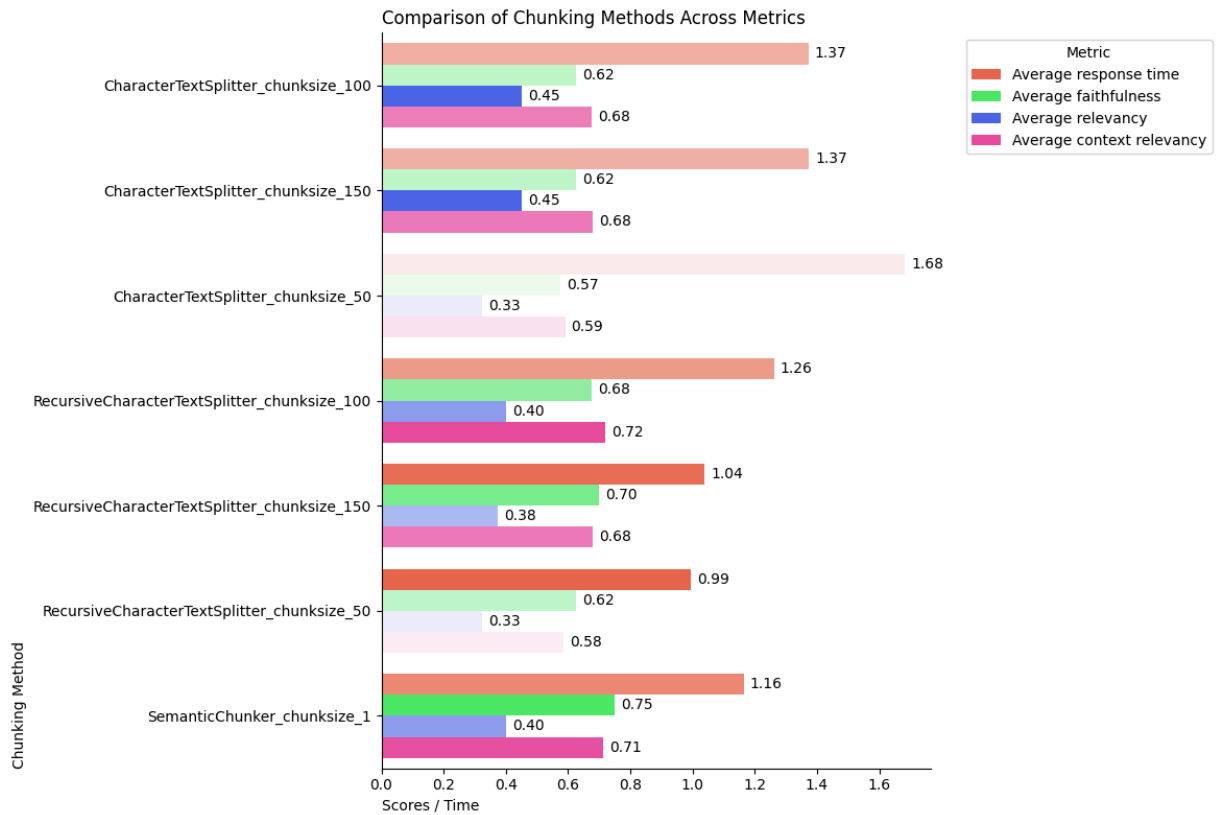
plt.figure(figsize=(12, 8))
bars = sns.barplot(data=melted_df,
                   x="Score",
                   y="Chunking Method",
                   hue="Metric",
                   palette=metric_colors)

# Annotate each bar with its value
for bar, row in zip(bars.patches, melted_df.itertuples()):
    bar_x = bar.get_width()
    bar_y = bar.get_y() + bar.get_height() / 2
    plt.text(bar_x + 0.02,
             bar_y,
             f"{row.Score:.2f}",
             color='black',
             va='center',
             fontsize=10)

for bar, opacity in zip(bars.patches, melted_df['Adjusted Opacity']):
    bar.set_alpha(opacity)

plt.xlabel("Scores / Time", loc="left")
plt.ylabel("Chunking Method", loc="bottom")
plt.title("Comparison of Chunking Methods Across Metrics", loc='left')
plt.legend(title="Metric", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
sns.despine()
plt.show()

```



Key Observations:

Average Response Time:

- The RecursiveCharacterTextSplitter_chunksize_50 has the lowest response time (0.993778 seconds), indicating it is the fastest method.
- The CharacterTextSplitter_chunksize_50 has the highest response time (1.682769 seconds), making it the slowest.

Average Faithfulness:

- The SemanticChunker_chunksize_1 achieves the highest faithfulness score (0.750), meaning it provides the most factually accurate responses.
- The CharacterTextSplitter_chunksize_50 has the lowest faithfulness score (0.575), indicating it struggles with factual accuracy.

Average Relevancy:

- All methods have relatively low relevancy scores, with the CharacterTextSplitter_chunksize_100 and CharacterTextSplitter_chunksize_150 tied at 0.450.
- The CharacterTextSplitter_chunksize_50 and RecursiveCharacterTextSplitter_chunksize_50 have the lowest relevancy scores (0.325), suggesting they are less aligned with the query intent.

Average Context Relevancy:

- The RecursiveCharacterTextSplitter_chunksize_100 achieves the highest context relevancy score (0.718750), meaning it retrieves the most relevant chunks for the query.
- The CharacterTextSplitter_chunksize_50 has the lowest context relevancy score (0.590625), indicating it retrieves less relevant chunks.

Chunking Method Use Case Recommendation

- SemanticChunker_chunksize_1 Best for high accuracy and meaningful context retrieval.
- RecursiveCharacterTextSplitter_chunksize_100 Best for high context relevancy with good accuracy.
- RecursiveCharacterTextSplitter_chunksize_50 Best for speed-critical tasks with less focus on accuracy.
- CharacterTextSplitter_chunksize_50 Avoid due to poor performance across all metrics.

We can see the result in other way, for example by a heat map, where we can see how were the metrics per question and chunking method, for example the **context relevancy score**.

```
In [ ]: import os
import json
import pandas as pd

eval_dir = './chunks_evaluations'
all_dfs = []

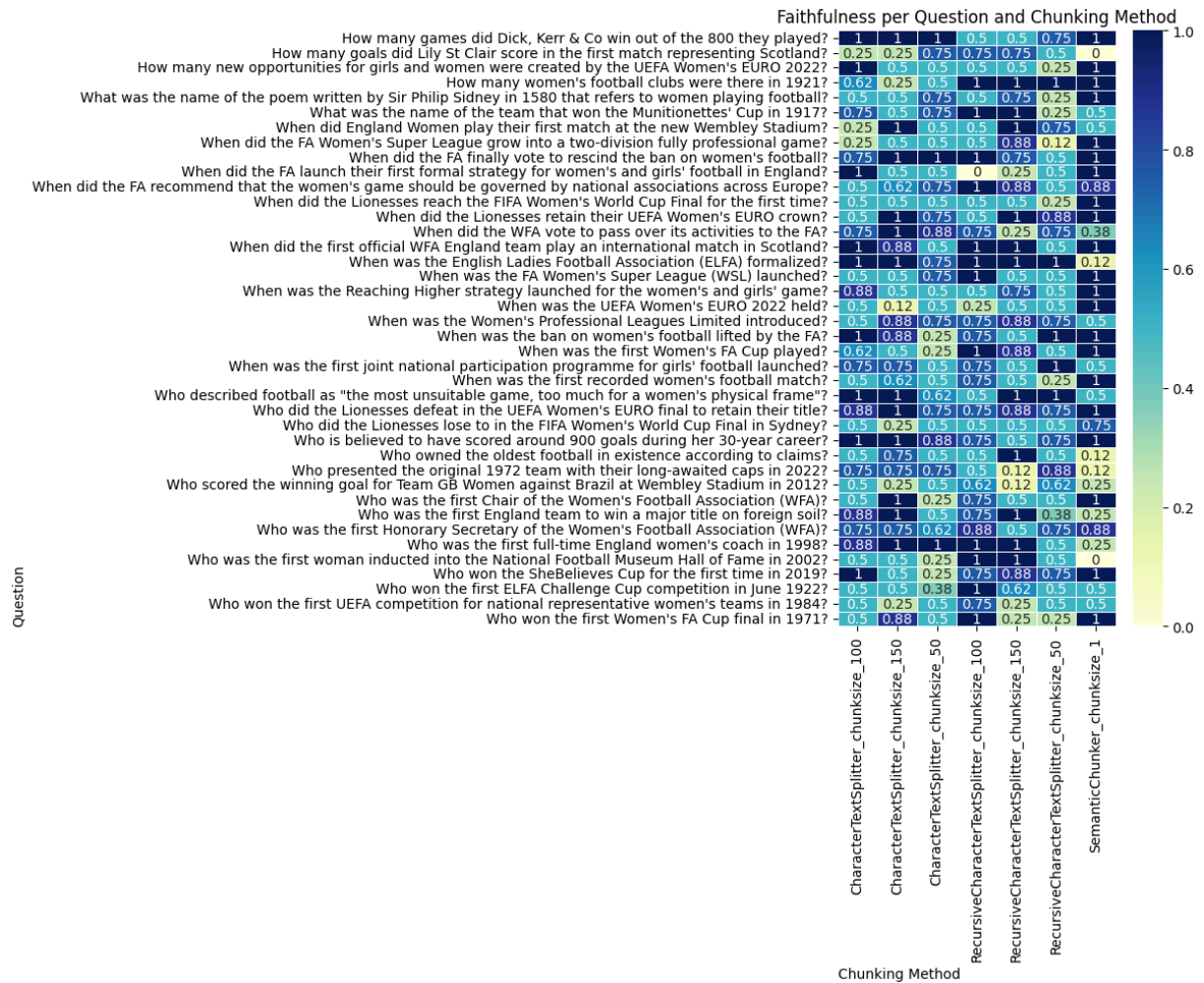
for filename in os.listdir(eval_dir):
    if filename.endswith('.json'):
        method = filename.replace('_evaluation_answers.json', '')
        with open(os.path.join(eval_dir, filename), 'r', encoding='utf-8') as f:
            data = json.load(f)
            df = pd.DataFrame(data)
            df['method'] = method
            all_dfs.append(df)

# Combine all methods into one DataFrame
full_df = pd.concat(all_dfs, ignore_index=True)
```

```
In [236... full_df['order'] = full_df['question'].str.split('.').str[0].str.strip()
full_df['question'] = full_df['question'].str.replace(r'^\d+\.\s*', '', regex=True)

# Recreate the heatmap DataFrame
heatmap_df = full_df.pivot(index='question', columns='method', values='context_rele

plt.figure(figsize=(12, 10))
sns.heatmap(heatmap_df, annot=True, cmap='YlGnBu', linewidths=.5)
plt.title('Faithfulness per Question and Chunking Method')
plt.ylabel('Question', loc='bottom')
plt.xlabel('Chunking Method', loc='left')
plt.tight_layout()
plt.show()
```

From above we can see that the **Semantic Chunker** consistently achieves high scores across most questions, indicating its ability to provide contextually relevant and accurate chunks. Also, the **Recursive Character Text Splitter** with a chunk size of 100 also performs well, particularly in terms of context relevancy. It is a good choice for balancing accuracy and retrieval quality. We can further check what happen if we increase the chunk size to 150 and 200.

Conclusion

This notebook explores various chunking techniques and evaluates their effectiveness in the context of Retrieval-Augmented Generation (RAG) pipelines. The key findings and insights are summarized below:

1. Chunking Techniques:

- Different chunking methods, including Fixed-Size Chunking, Recursive Chunking, Semantic Splitting, and LLM-Based Chunking, were implemented and compared.
- Each method has its strengths and weaknesses, with trade-offs between speed, semantic coherence, and accuracy.

2. **Evaluation Metrics:**

- The evaluation focused on metrics such as Faithfulness, Relevancy, Context Relevancy, and Response Time.
- These metrics provided a comprehensive understanding of how well each chunking method supports accurate and relevant responses in RAG pipelines.

3. **Key Observations:**

- SemanticChunker_chunksize_1 achieved the highest faithfulness score, making it the most accurate method for retrieving meaningful context.
- RecursiveCharacterTextSplitter_chunksize_100 demonstrated the best context relevancy, indicating its ability to retrieve highly relevant chunks for queries.
- RecursiveCharacterTextSplitter_chunksize_50 was the fastest method, making it suitable for speed-critical tasks.
- CharacterTextSplitter_chunksize_50 performed poorly across most metrics and is not recommended for use.

4. Recommendations:

- Use Semantic Chunking for tasks requiring high accuracy and meaningful context retrieval.
- Opt for Recursive Chunking when balancing context relevancy and accuracy is critical.
- Avoid Fixed-Size Chunking for tasks requiring semantic coherence, as it often splits text mid-sentence or mid-thought.

5. Visualization:

- The heatmap and bar plots provided clear visual insights into the performance of each method, aiding in the selection of the most appropriate chunking strategy for specific use cases.